Modeling Foreign Exchange Risk using GARCH-Type Models and Value-at-Risk

Research Question

Can the GARCH model and its extensions improve the estimation of market risk for major EUR-denominated currency pairs compared to a simple historical volatility approach, and to what extent does the choice of conditional variance specification and innovation distribution affect Value-at-Risk forecast accuracy?

Data

Daily exchange rate quotations obtained from Yahoo Finance for three EUR-denominated currency pairs: EUR/USD, EUR/GBP, and EUR/CHF.

Methodology

1) Transform exchange rate prices into daily log-returns;
2) Estimate three families of conditional variance models - symmetric GARCH(1,1), exponential GARCH (EGARCH(1,1)), and Glosten-Jagannathan-Runkle GARCH (GJR-GARCH(1,1)) — to capture time-varying volatility, with EGARCH and GJR-GARCH specifically designed to capture asymmetric responses of volatility to positive and negative shocks (the leverage effect);
3) For each conditional variance model, estimate three alternative innovation distributions - Gaussian (normal), Student-t, and skewed Student-t - yielding nine model–distribution combinations per currency pair;
4) Compute Value-at-Risk at the 95% and 99% confidence levels using the corresponding distributional quantiles;
5) Apply an out-of-sample forecasting framework using rolling window estimation, where each model is repeatedly re-estimated on a fixed-size moving sample;
6) Evaluate forecast performance through VaR backtesting, comparing actual violations with the expected violation rate using the Kupiec unconditional coverage test, the Christoffersen conditional coverage test, and the dynamic quantile (DQ) test;
7) Rank competing specifications using information criteria (AIC, BIC), log-likelihood values, and the backtesting outcomes.

Comparative Analysis

The study performs a two-dimensional comparison. The first dimension contrasts Historical VaR with all parametric GARCH-based VaR estimates, assessing whether modeling conditional volatility improves risk estimation relative to a non-parametric historical approach. The second dimension is internal to the parametric framework: it compares symmetric GARCH against asymmetric extensions (EGARCH, GJR-GARCH), and across these models compares the three innovation distributions (Gaussian, Student-t, skewed Student-t). The objective is to identify whether the gains in VaR accuracy come predominantly from (i) modeling time-varying volatility, (ii) accounting for asymmetric volatility responses, or (iii) using fat-tailed and skewed innovation distributions.

Model–Distribution Comparison

Table 1. Model–Distribution Specifications and Captured Features

Model	Innovation Distribution	Captures Asymmetry (Leverage)	Captures Heavy Tails	Captures Skewness
GARCH(1,1)	Gaussian	No	No	No
GARCH(1,1)	Student-t	No	Yes	No
GARCH(1,1)	Skewed Student-t	No	Yes	Yes
EGARCH(1,1)	Gaussian	Yes	No	No
EGARCH(1,1)	Student-t	Yes	Yes	No
EGARCH(1,1)	Skewed Student-t	Yes	Yes	Yes
GJR-GARCH(1,1)	Gaussian	Yes	No	No
GJR-GARCH(1,1)	Student-t	Yes	Yes	No
GJR-GARCH(1,1)	Skewed Student-t	Yes	Yes	Yes

Table 1 summarizes the nine model–distribution combinations considered in this study and the stylized features of FX returns that each combination is theoretically able to capture.

Expected Empirical Results

- Figure 1. Daily Log-Returns of EUR/USD, EUR/GBP, and EUR/CHF;
- Figure 2. Squared Returns as a Volatility Proxy across the three currency pairs;
- Figure 3. Estimated Conditional Volatility from GARCH(1,1), EGARCH(1,1), and GJR-GARCH(1,1);
- Table 2. Estimated Parameters for GARCH, EGARCH, and GJR-GARCH under Gaussian, Student-t, and skewed Student-t innovations;
- Table 3. Goodness-of-Fit Comparison (log-likelihood, AIC, BIC) across all model–distribution combinations;
- Figure 5. Value-at-Risk Forecasts at the 95% and 99% confidence levels for each currency pair;
- Table 4. VaR Backtesting Results, including unconditional coverage (Kupiec), conditional coverage (Christoffersen), and dynamic quantile (DQ) tests.